

Who Gets Caught: Blockholder Disclosure Rules and Market Inference

Austin Li

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Abstract

A blockholder disclosure rule has two dials, a stake threshold and a filing clock. Together they split every possible history into a flagged cell, in which the filing has landed, and a pooled cell, in which the market reads order flow alone. Disclosure regulation changes what the market learns from silence. Non-disclosure itself alters the market's inference. At fixed trading policies, the liquidity sensitivity of the expected engagement premium factorises as the pooled share times the pooled cell's own sensitivity. The factorisation and the liquidity invariance of the flagged endpoint are PROVED. Tightening the threshold lowers sensitivity through a PROVED weight effect. Its composition effect is NUMERICAL on the order-size-two, $H = 10$, $T = 5$ grid at frozen benchmark policy, for four adjacent threshold pairs and κ from 0.15 to 0.85 in steps of 0.01. Shortening the clock lowers sensitivity exactly when the clock's weight and composition ratios have product at most one, a PROVED equivalence. A PROVED corollary characterises the composition ratio by the noise sensitivity of the premium mass the shorter clock removes from the pool, including the re-pricing of survivors. Under a common nonzero sign, $C_T \leq 1$ exactly when the caught leg is at least as noise-sensitive as the pool and no more than $((2-\varphi)/\varphi)$ times as sensitive, where φ is the share caught. The empirical measurement is descriptive, with the post-minus-pre differences labelled ESTIMATED. Among initial Schedule 13D campaigns with readable stakes from 2021 to 2025, the median stake at filing is 13.1 percent of class before the February 2024 clock change and 12.8 percent after. The share filed within five business days rises from 32.7 to 75.9 percent.

1 Introduction

A blockholder who learns that a company is worth more than the market price can sell, wait, or try to change the company. If she builds a large enough position while she waits, the law makes her disclose it. In the United States, Section 13(d) of the Williams Act requires beneficial owners of more than 5 percent of a registered equity class who do not qualify for or elect Schedule 13G to file a Schedule 13D. Until February 2024 the initial deadline was ten calendar days; it is now five business days (U.S. Securities and Exchange Commission, 2023). That rule gives two policy variables: the threshold determines which positions become public, and the clock determines how much trading can take place before the filing. The central thesis of this paper is that disclosure regulation changes what the market learns from silence. A disclosure rule releases information on filed histories and selects which informed histories leave the unfiled pool. The resulting pool changes the market’s inference when no filing lands.

The disclosure rule divides all histories into a flagged cell and a pooled cell. After a filing, Item 4 disclosures make public the blockholder’s accumulated stake, intended plans, and engagement intent, represented in the model by public disclosure of the stake, residual target order, and engagement indicator. Before a filing lands, the market observes order flow alone and must infer whether an informed holder is trading in secret. The formal outcome is the liquidity sensitivity of the engagement-related component of the expected takeover premium, $\Delta^{\text{act}} = \Delta_m \mathbb{E}[\pi p]$, which governs a potential bidder’s takeover decision and omits the baseline entry term $m_0 \mathbb{E}[p]$. Liquidity changes inference in the pooled cell but not what the market learns from a completed filing. At fixed trading policies, the flagged posterior, the flagged price, the bidder’s entry probability, and the flagged cell’s average premium do not depend on the noise-trading intensity κ .

The first result is the factorisation $\mathcal{S} = (1 - \Omega)\mathcal{S}_P$, where Ω is the flagged probability and $\mathcal{S}_P$ is the pooled cell’s own noise sensitivity. The factorisation and the κ -invariance of the flagged cell are PROVED. All theoretical results hold trading policies fixed at the benchmark profile. This isolates changes in market information from the blockholder’s endogenous policy response. The two dials then operate through distinct mechanisms. Tightening the threshold weakly raises Ω , so its weight effect attenuates noise sensitivity by proof. Its composition effect is NUMERICAL on a five-node ladder of the stake-at-filing distribution, $\tau \in [0.0924, 0.0970]$, under order size two, $H = 10$, $T = 5$, at frozen benchmark policy, with κ from 0.15 to 0.85 in steps of 0.01. On that grid the composition ratio C_τ is at most one at all four adjacent threshold pairs, and $W_\tau C_\tau$ is at most 0.772. For the clock, the model measures the window margin T in discrete trading rounds. A PROVED theorem establishes that a shorter window lowers noise sensitivity if and only if $W_T C_T \leq 1$. A PROVED corollary then characterises who gets caught. Write φ for the share of the long-clock pool removed by the shorter clock and s_B for the sensitivity of the premium mass removed, including the repricing of survivors. Under a common sign ($s_A s_B > 0$), the composition ratio satisfies $C_T \leq 1$ if and only if s_B lies weakly between s_A and $((2 - \varphi)/\varphi)s_A$, and the shorter clock lowers sensitivity overall, $W_T C_T \leq 1$, if and only if s_B lies weakly between 0 and $(2/\varphi)s_A$.

At order size two, a building purchase has the size of two noise lumps. Flow then has five values. Only $+\bar{z}$ is ambiguous, arising either from a building purchase with negative noise or from no building purchase with positive noise. The pooled experiment at higher κ is a garbling of the experiment at lower κ in the sense of Blackwell (1953). The result is PROVED. It gives a Markov kernel and an exact polynomial form for the pooled premium. The convexity-signed ordering does

not require normality. Section 3 gives the rationale for the order-size choice.

The registered empirical analysis gives institutional context for the stake and clock parameters. It does not measure market inference directly. The population is initial Schedule 13D campaigns from 2021 to 2025, taking the campaign (one subject firm and trigger date) as the unit, without screening by Item 4 text, and reading the maximum percentage across reporting persons. Stakes are readable for 91.6 percent of campaigns. In that sample, the post-minus-pre differences in the stake at filing B^F are labelled ESTIMATED. The share of campaigns filed within five business days rises from 32.7 percent before the February 2024 clock change to 75.9 percent after, and the median trigger-to-filing delay falls from 7.0 to 5.0 business days. The comparison has no control group and identifies no causal effect.

Standard discrete-order trading models with exogenous disclosure typically fix the composition of the unfiled pool, so the question of its sign does not arise. This paper analyses the clock dial result and the who-gets-caught characterisation. The exit-and-liquidity literature studies whether liquidity finances monitoring, and the disclosure literature studies the level of the threshold; neither separates the statutory rule into two distinct margins acting on a partition of histories. Section 2 sets out both literatures and the institutional facts. Section 3 states the model. Section 4 states the results, each with its label. Section 5 reads them against the calibration. Section 6 presents the empirics, and Section 7 closes. The proofs and the hypothesis lists are in the appendix.

2 Institutional setting and related literature

2.1 The disclosure rule and its two margins

Section 13(d) requires beneficial owners of more than five percent of a registered equity class to report their holdings. Investors with an activist intent or who do not qualify to report on Schedule 13G must file Schedule 13D, disclosing their stake and purpose. The statutory initial filing window was ten calendar days until the SEC amendments effective February 5, 2024 shortened it to five business days (U.S. Securities and Exchange Commission, 2023). The threshold determines which holdings trigger a filing. The clock determines how long a crossed position can remain in the pooled cell. Other disclosure systems also vary these two margins, so the distinction is not tied to one statutory change.

The theoretical model abstracts this institution into two margins, defining the window margin T in discrete trading rounds. The clock starts when the stake reaches τ , and the filing lands T trading rounds later. The numerical comparison between $T = 10$ and $T = 5$ below is a comparative static across trading-round windows in the model, rather than a calibrated replica of the calendar-day to business-day reform. Filing early is outside the model. The pooled cell describes information available to the market makers who set prices. Other parties may learn about accumulation through separate channels before a filing, as Zeng (2026) documents for target managers and stock-surveillance vendors. That possibility does not change the public filing partition studied here.

2.2 Exit, voice, and liquidity

The classic trade-off originates with Hirschman (1970). Coffee (1991) and Bhidé (1993) argue that liquidity weakens governance by letting blockholders sell instead of fight; Maug (1998) reverses the logic. Liquid markets let a monitor recoup the costs of engagement through informed trading, and with an endogenous stake liquidity raises monitoring. Kahn and Winton (1998) show the sign of the trading motive depends on whether engagement deepens the blockholder’s information advantage; Faure-Grimaud and Gromb (2004) make price informativeness and insider incentives complements; Admati and Pfleiderer (2009) make exit itself a disciplining threat. On the empirical side, Edmans, Fang, et al. (2013) find more liquid targets shift from voice to exit, and Norli et al. (2015) find liquidity predicts the incidence of costly voice.

This literature uses different meanings of liquidity, so its comparative statics do not map to one common question. It also focuses on monitoring and trading rather than the disclosure partition faced by a bidder. The formal object analysed here is the liquidity sensitivity of the engagement-related component of the expected takeover premium, $\Delta^{\text{act}} = \Delta_m \mathbb{E}[\pi p]$, when the legal rule sorts histories into flagged and pooled cells. It omits the baseline entry term $m_0 \mathbb{E}[p]$, which can move with liquidity when entry moves. The factorisation and comparative statics apply specifically to this engagement component rather than to total takeover premia or shareholder welfare.

2.3 Trading, disclosure, and the market for control

The trading model uses discrete order flow from the Kyle tradition (Kyle, 1985; Glosten and Milgrom, 1985), following the formulation in Edmans, Goldstein, et al. (2015). It also uses the feedback between prices and real decisions (Edmans, Goldstein, et al., 2012; Bond et al., 2012; Goldstein, 2023). Grossman and Hart (1980) make the takeover premium the welfare object for dispersed minority shareholders, and Burkart, Gromb, et al. (1998) show why higher premia protect them.

Several theory papers study related information splits. In Kyle and Vila (1991), pooling comes from the trader’s mixing rather than a legal rule. Collin-Dufresne and Fos (2015) and Collin-Dufresne and Fos (2016) study informed blockholder trading when the trader’s presence is known. Back et al. (2018) has a fixed disclosure horizon in a continuous-time Kyle model, where a shorter horizon changes the scale of noise-trading volatility. The legal threshold and clock here instead change which histories the market pools before a bidder acts.

Cetemen et al. (2026) studies how blockholders time trades and coordinate through market signals. Corum (2025) limits how much of a stake can be traded before disclosure but does not separate the threshold from the clock. Corum and Levit (2019) and Ordóñez-Calafí and Bernhardt (2022) study disclosure thresholds without a distinct window margin. Burkart and Lee (2022) studies the takeover premium while leaving anonymous pre-disclosure toehold accumulation outside its model.

2.4 Structural and empirical work on activism

Structural work estimates campaign costs, reputation, disclosure form, and takeover outcomes. Gantchev (2013) estimates the sequential costs of campaigns. Johnson and Swem (2021) models

reputation when buying and filing occur together. Albuquerque et al. (2022) studies the 13D-versus-13G choice with announcement returns. Celentano and Levine (2025) models activism, takeovers, and managerial discipline with a fixed disclosure threshold. These models do not separate a threshold margin from a filing clock. Empirical work links engagement campaigns to acquisitions and returns (Brav et al., 2008; Greenwood and Schor, 2009; Becht et al., 2017). Zeng (2026) measures information that target managers receive before the public filing.

Recent work studies the clock change directly. Polk et al. (2024) uses pre-rule variation in filing delays, while Trivedi (2026) compares filings around the new rule. The empirical measurement reports the stake at filing and realised filing delays before and after the statutory change. The theory studies the threshold and clock as separate fixed-policy changes to market inference.

3 The model

3.1 The object and the partition

The disclosure rule divides the market’s information into two cases. A stake threshold τ and a window margin T determine whether the filing has arrived by the control decision. In the *flagged* cell, the market sees the stake and the intention to engage. In the *pooled* cell, it sees only order flow and must infer whether an informed blockholder is trading. The cells are exhaustive. The outcome of interest is the expected engagement-related premium $\Delta^{\text{act}}(\kappa, \tau, T)$, its two cell averages M_F and M_P , and their liquidity sensitivities. A lower threshold is a tighter threshold margin; a shorter window is a tighter window margin.

The model uses the discrete order-flow structure of Kyle (1985) and Edmans, Goldstein, et al. (2015), competitive prices as in Glosten and Milgrom (1985), and the takeover premium as the dispersed shareholder’s control-outcome measure in the spirit of Grossman and Hart (1980). The model adds the legal partition to these elements.

3.2 Timing

Trading dates are $d = 0, \dots, H$. Nature draws the standalone value and the blockholder’s signal. A cutoff vector then maps the signal into one of three plans. Exit sells the initial stake on date zero and holds nothing thereafter. Hold keeps the initial stake. Voice builds toward a signal-dependent target and carries the engagement indicator $a = 1$. Only Voice can cross the disclosure threshold.

On each date, a binary order mark records whether the Voice path schedules a building purchase. The strategic order attached to that mark and the independent noise order form observed flow. Market makers set the date’s pooled price from the flow history and the fact that no filing has yet landed. The blockholder pays the pooled price for the scheduled change in the stake. The path does not respond to realised flow or prices.

For a Voice path, the clock starts at the first date c on which the stake reaches τ , and the filing date is $f = c + T$ in trading rounds when $f \leq H$. The within-date order is explicit. The blockholder pays the corresponding pooled price for every scheduled purchase through date f , including the purchase that determines $B^F = B(s, f)$. The filing then lands. The blockholder pays the flagged price for the remaining purchase $Q^F = b^*(s) - B^F$. The bidder acts after this update. If no filing lands by H , the bidder acts on the pooled control-node history.

3.3 Primitives

The firm's standalone value is $v \sim N(\mu_v, \sigma_v^2)$. The blockholder observes $s = v + \varepsilon$, where $\varepsilon \sim N(0, \sigma_\varepsilon^2)$ is independent of v . Thus $s \sim N(\mu_v, \sigma_s^2)$, with $\sigma_s^2 = \sigma_v^2 + \sigma_\varepsilon^2$, and

$$\mathbb{E}[v | s] = \mu_v + \beta(s - \mu_v), \quad \beta = \frac{\sigma_v^2}{\sigma_v^2 + \sigma_\varepsilon^2}.$$

The analytic model uses the full signal line. The computations retain $s \in [\mu_v - 6\sigma_s, \mu_v + 6\sigma_s]$ and renormalise the truncated Gaussian law on that interval. A potential bidder has an independent shock $\xi \sim N(0, \sigma_\xi^2)$, mean synergy $\bar{S}$, and entry cost $K > 0$. Engagement adds $\Delta_V \geq 0$ to standalone value and changes the takeover premium from m_0 to m_1 , where $\Delta_m = m_1 - m_0 > 0$. The restriction $m_0 \geq 0$ gives a unique continuous pricing root (Lemma 8 in the appendix).

The Voice target follows the strictly increasing algebraic sigmoid used in the computations,

$$b^*(s) = b_0 + (\bar{b} - b_0)G\left(\frac{s - \mu_v}{\sigma_s}\right), \quad G(x) = \frac{1 + x/\sqrt{1 + x^2}}{2}.$$

Its accumulation length is

$$n(s) = \text{clip}\left(\left\lceil n_{\text{scale}}(H + 1) \left[1 - G\left(\frac{s - \mu_v}{\sigma_s}\right)\right]\right\rceil, 1, H + 1\right),$$

which is weakly decreasing in s . The Voice stake path is

$$B_V(s, d) = b_0 + (b^*(s) - b_0) \min\left\{1, \frac{d + 1}{n(s)}\right\}.$$

Exit has $B_E(s, -1) = b_0$ and $B_E(s, d) = 0$ from date zero onward. Hold keeps $B_H = b_0$. The threshold satisfies $b_0 < \tau$. The first crossing is $c_j(s; \tau) = \inf\{d : B_j(s, d) \geq \tau\}$, with value $+\infty$ if there is none, and $D = \mathbf{1}\{a = 1, c + T \leq H\}$.

The order mark and order size are distinct. The mark

$$g_{jd}(s) = \Gamma(B_j(s, d) - B_j(s, d - 1)) \in \{0, 1\}, \quad \Gamma(x) = \mathbf{1}\{x \geq \bar{\gamma}\},$$

is a building-purchase indicator. Its strategic order has size two noise lumps, $q_{jd}(s) = 2\bar{z}g_{jd}(s)$. Independent noise takes the values $-\bar{z}$, 0, and $+\bar{z}$, with probabilities $\kappa/2$, $1 - \kappa$, and $\kappa/2$. Noise orders are independent across dates and independent of the primitive draws. Observed pooled flow is $X_d = q_{jd} + z_d$. Higher κ is a noisier, more liquid market.

Under Exit, the blockholder sells the initial stake at the pooled date-zero price but leaves no order mark. This is a model assumption. The public statistic records buy-side stake building, not all signed inventory changes, and therefore keeps Exit and Hold in the same mark class. Modeling signed liquidation would add a third strategic-order support and create a different information experiment from the five-flow experiment studied here.

A pooled history records $(X_0, \dots, X_d)$ and whether the filing has landed. A filing reports $F = (B^F, a = 1)$, and the flagged tuple adds the residual order, $S_F = (B^F, Q^F, a = 1)$. Institutionally, Item 4 of Schedule 13D requires disclosing the purpose of the acquisition and any plans or proposals

regarding future purchases. In the model, the public tuple $\mathbf{S}_F$ captures this requirement: the filing reveals the accumulated stake B^F , the activist's scheduled remaining purchase $Q^F = b^*(s) - B^F$ toward her target, and the engagement commitment $a = 1$. This is a stronger information set than a filing alone: Item 4 motivates the disclosure of plans, and the model takes the residual order and the engagement commitment as exactly disclosed, which a filing does not guarantee. At the control node, public information is the pooled history if $D = 0$ and $\mathbf{S}_F$ if $D = 1$. The bidder's shock remains private. The engagement posterior is $\pi(\mathcal{I}) = \mathbb{P}(a = 1 \mid \mathcal{I})$ and equals one after a filing. The bidder enters with probability

$$p(\mathcal{I}) = 1 - \Phi\left(\frac{P + K + m_0 + \pi\Delta_m - \bar{S}}{\sigma_\xi}\right). \quad (1)$$

With $\mathbf{B}$ the entry indicator, the terminal shareholder payoff is

$$Y = (1 - \mathbf{B})(v + a\Delta_V) + \mathbf{B}(P + m_0 + a\Delta_m), \quad (2)$$

and the competitive control-node price solves $P(\mathcal{I}) = \mathbb{E}[Y \mid \mathcal{I}]$. Earlier pooled prices are tower expectations of the solved control-node payoff. Positive-probability histories use Bayes' rule. The pricing map assigns a fixed full-support reference belief to an empty mark-path history.

For plan j , the blockholder evaluates

$$U_j(s) = \mathbb{E}\left[b_j^*(s)Y - C_j^{\text{trade}} - a_j C_j(s) \mid s, j\right].$$

The trading cost sums each position change times the price specified by the within-date timing. The engagement cost is positive for Voice and zero for Exit and Hold. A weakly ordered cutoff vector maps the signal into Exit, Hold, or Voice. The calibration uses the resulting cutoff vector as its baseline policy and freezes it across the comparative statics. The blockholder executes the selected path without responding to realised prices or flow. The finite menu, bounded premium kernel, legal-clock rules, and single probability space are listed as (S1) to (S11) in the appendix and apply to the results below.

3.4 The order-size choice

A stake-building order should be large enough to move outside an ordinary noise order, but not so large that its sign reveals the blockholder in every round. Two noise lumps are a simple middle scale. A building purchase is twice the size of one liquidity order, yet a noise sale can still offset half of it and produce the same net flow as an idle blockholder facing a noise buy. The model preserves one camouflage event while most flows reveal whether stake building occurred.

Flow takes the five values $\{-\bar{z}, 0, \bar{z}, 2\bar{z}, 3\bar{z}\}$. A building date produces $\{\bar{z}, 2\bar{z}, 3\bar{z}\}$ and an idle date produces $\{-\bar{z}, 0, \bar{z}\}$. The supports meet only at $\bar{z}$. This single overlap makes higher liquidity a garbling of lower liquidity and gives the pooled premium a polynomial form in κ .

3.5 Fixed policies and the two dials

Every theoretical comparison holds the plan menu, cutoff vector, and execution paths fixed while the disclosure rule (τ, T) moves. Each comparison holds κ fixed across the two rules. The numerics hold

the solver's baseline policy fixed throughout. These comparisons isolate the market's information problem rather than a policy response by the blockholder. The formal outcome is the engagement-related component of the expected takeover premium,

$$\Delta^{\text{act}} = \Delta_m \mathbb{E}[h(\mathcal{I}_H)], \quad h(\mathcal{I}) = \pi(\mathcal{I}) p(\mathcal{I}),$$

omitting the baseline entry term $m_0 \mathbb{E}[p]$. The cell averages are $M_F = \Delta_m \mathbb{E}[h \mid D = 1]$ and $M_P = \Delta_m \mathbb{E}[h \mid D = 0]$ with D the disclosure indicator, the flagged weight is $\Omega = \mathbb{P}(D = 1)$, and the sensitivities are $\mathcal{S} = |\partial_\kappa \Delta^{\text{act}}|$ and $\mathcal{S}_P = |\partial_\kappa M_P|$. The two dials act through the weight legs

$$W_\tau = \frac{1 - \Omega(\tau', T)}{1 - \Omega(\tau, T)} \quad (\tau' < \tau), \quad W_T = \frac{1 - \Omega(\tau, T')}{1 - \Omega(\tau, T)} \quad (T' < T),$$

and the composition legs $C_\tau = \mathcal{S}_P(\tau', T)/\mathcal{S}_P(\tau, T)$ and $C_T = \mathcal{S}_P(\tau, T')/\mathcal{S}_P(\tau, T)$, both unsigned.

By Lemma 9 in the appendix, under fixed policies, because Voice stakes are weakly increasing in the calendar date and crossing dates are unchanged, shortening the window margin lands the filing weakly earlier on every path flagged under both windows, and on those paths the stake at filing B^F weakly falls. The shorter window can also flag paths that the longer one leaves in the pool, so the set of filers can change as well. The empirical section states this pathwise monotonicity next to the measured stakes and draws nothing causal from the pairing.

4 Results at fixed policies

Each statement reports its status as PROVED, NUMERICAL, or ESTIMATED. PROVED marks an analytic result. NUMERICAL marks a result on the grid named at the point of claim. ESTIMATED marks a descriptive estimate with its design and uncertainty. The full hypothesis lists (S1) to (S11) and (C-1) to (C-3) are in the appendix.

4.1 The partition and the factorisation

LABEL: PROVED.

Lemma 1 (Disclosure partition; Appendix Section 1.1, Lemma 2). *Under the standing conditions of Section 3, the disclosure indicator $D = \mathbf{1}\{a = 1, c(\tau) + T \leq H\}$ is a measurable function of the control-node history, and the flagged cell $\mathcal{C}_F = \{D = 1\}$ and the pooled cell $\mathcal{C}_P = \{D = 0\}$ are disjoint, cover the space, and are read off the observed history. No restriction on the mass of either cell is used: Ω may be 0 or 1, and then the empty-weight cell is null, possibly empty.*

LABEL: PROVED.

Lemma 2 (Two-cell decomposition; Appendix Section 1.1, Lemma 3). *Under the same conditions, with the partition of Lemma 1 and whenever $0 < \Omega < 1$,*

$$\Delta^{\text{act}} = \Omega M_F + (1 - \Omega) M_P.$$

At $\Omega = 1$ the identity reads $\Delta^{\text{act}} = M_F$ and at $\Omega = 0$ it reads $\Delta^{\text{act}} = M_P$; in each of those cases the average over the null cell is undefined rather than zero, and it is not determined by the law.

LABEL: PROVED.

Proposition 1 (Factorisation of the liquidity sensitivity; Appendix Section 1.1, Proposition 1). *Assume the standing conditions, an interior partition $0 < \Omega < 1$ at the policy under consideration, the invariance of the flagged endpoint in κ at fixed policies (Lemma 3), and differentiability of $\kappa \mapsto M_P$. Then*

$$\mathcal{S}(\kappa, \tau, T) = (1 - \Omega(\tau, T)) \mathcal{S}_P(\kappa, \tau, T) \quad (3)$$

exactly, and the same factor scales the total variation of Δ^{act} over any grid in κ , where no differentiability is used. At order size two the differentiability hypothesis is automatic: the closed form of Lemma 4 exhibits M_P as a polynomial in κ .

LABEL: PROVED.

Lemma 3 (The flagged cell does not move with liquidity; Appendix Section 1.1, Lemma 1). *Fix the cutoff policy and the execution policy, and let the disclosure rule be fixed. Under the standing conditions, with the composed terminal target strictly increasing on the flagged signal region almost surely, uniqueness of the inner pricing fixed point, $\Omega > 0$, and a bidder entry rule whose probability given the public information set neither depends on v beyond it nor moves the entry shock, the flagged tuple $\mathbf{S}_F$ makes the pre-filing pooled history conditionally independent of the primitive triple (v, s, ξ) , and the flagged posterior, the flagged price, the entry probability and M_F do not depend on κ . In addition Ω does not depend on κ at fixed policies.*

The appendix gives all four proofs. The partition is a measurable split of the history space. The decomposition integrates a bounded kernel over its two cells. The factorisation then uses the fact that the flagged endpoint and the flagged weight are constant in κ . The invariance proof uses the flagged tuple to recover the signal and its posterior. The pre-filing pooled price can still move with κ , and all four statements hold at fixed policies.

4.2 The pooled experiment ordered by liquidity

LABEL: PROVED.

Lemma 4 (Liquidity garbles the pooled experiment, and a closed form; Appendix Section 2, Lemmas 5–7). *Fix a policy with $\Omega(\tau, T) < 1$ and $\kappa \in (0, 1)$ at order size two. Write R for the set of rounds whose flow differs from one lump $\bar{z}$.*

- (a) *The value $\bar{z}$ carries likelihood ratio one across types, so R is independent of the blockholder's type, and on R the flow determines the blockholder's order mark. The pooled posterior after any pooled history is therefore the pooled type law conditioned on the marks the history reveals, and nothing else.*
- (b) *For $0 < \kappa < \kappa' < 1$ there is a Markov kernel, not depending on the type, that carries the law of every pooled history at κ into its law at κ' : the pooled experiment at κ' is a garbling of the pooled experiment at κ , in the sense of Blackwell (1953).*
- (c) *The pooled premium is exactly $M_P = \Delta_m \sum_S (1 - \varepsilon)^{|S|} \varepsilon^{H+1-|S|} \mathcal{G}(S)$ with $\varepsilon = \kappa/2$ and $\mathcal{G}(S)$ a number that does not depend on κ ; it is a polynomial in κ , differentiable on $(0, 1)$, with*

$\partial_\kappa M_P = -(\Delta_m/2) \sum_k (1 - \varepsilon)^k \varepsilon^{H-k} c_k$ and coefficients c_k that are closed forms in the pooled type weights. If the kernel's reduced form is concave on the pairs the pooled cell generates, every $c_k \leq 0$ and $\partial_\kappa M_P \geq 0$; if convex, all inequalities reverse.

The appendix constructs the kernel in part (b) by deleting revealed rounds and redrawing the rest. The pooled expectation of a convex or concave kernel is monotone in κ in the direction set by its curvature. This argument needs no normality assumption. Back et al. (2018) obtain a related ordering from a Gaussian mean-preserving spread of the market maker's posterior mean, while Ganuza and Penalva (2010) uses dispersion orderings of signal structures. The threshold composition result uses the closed form in part (c).

Condition 1 (Revelation dominance at a threshold pair). *Let $b_0 < \tau' < \tau$ at a common window T . The pair satisfies revelation dominance at κ if $|\mathcal{W}_{\tau',T}(\kappa)| \leq |\mathcal{W}_{\tau,T}(\kappa)|$, where $\mathcal{W}$ is the coefficient sum of Lemma 4(c). The condition states exactly that the composition ratio is at most one, $C_\tau \leq 1$: it is the theorem's conclusion restated on the closed form, not a restriction on primitives.*

4.3 The two dials

LABELS: PROVED AND NUMERICAL. Factorisation, the weight leg, and the closed-form implication below are PROVED. The composition conclusion $C_\tau \leq 1$ and the dial inequality are NUMERICAL on the order-size-two, $H = 10$, $T = 5$ grid at frozen benchmark policy, for all four adjacent threshold pairs and $\kappa \in [0.15, 0.85]$ in steps of 0.01.

Theorem 1 (The threshold dial at fixed policies; Appendix Section 2, Theorem 2). *Fix the plan and cutoff policies and a window T , and take $b_0 < \tau' < \tau$ with $0 < \Omega(\tau, T)$, $\Omega(\tau', T) < 1$ and $\mathcal{S}_P(\tau, T) > 0$. Assume the two-cell decomposition and the κ -invariance of the flagged average, each with its own hypotheses. Then:*

- (A) Factorisation. $\mathcal{S} = (1 - \Omega) \mathcal{S}_P$ exactly, at every $\kappa \in (0, 1)$.
- (B) Weight leg. $\mathcal{C}_F(\tau, T) \subseteq \mathcal{C}_F(\tau', T)$ and every newly flagged history is generated by a Voice plan, so $\Omega(\tau', T) \geq \Omega(\tau, T)$ and $W_\tau \in [0, 1]$. Equality holds when the threshold change reclassifies no history (almost surely).
- (C) Composition leg and the dial. Under Condition 1 at the pair and at κ , $C_\tau \leq 1$, and therefore

$$\frac{\mathcal{S}(\tau', T)}{\mathcal{S}(\tau, T)} = W_\tau C_\tau \leq 1. \quad (4)$$

LABEL: PROVED.

Theorem 2 (The clock dial at fixed policies; Appendix Section 1.1, Theorem 1). *Fix the liquidity κ , the threshold margin τ , and the plan and cutoff policies. Compare two window margins $T' < T$. Assume $b_0 < \tau$, that Voice stake paths are weakly increasing in the calendar date, that only Voice plans cross, $0 < \Omega(\tau, T) < 1$, $0 < \Omega(\tau, T') < 1$, $\mathcal{S}_P(\kappa, \tau, T) > 0$, and the hypotheses of Proposition 1 at both clocks, the differentiability of $\kappa \mapsto M_P$ each instance carries included. Then:*

- (i) the weight leg attenuates: $0 < W_T \leq 1$;

(ii) the sensitivity ratio factors as $\mathcal{S}(\kappa, \tau, T')/\mathcal{S}(\kappa, \tau, T) = W_T C_T$;

(iii) the shorter clock lowers noise sensitivity exactly when the product of the two legs is at most one,

$$\mathcal{S}(\kappa, \tau, T') \leq \mathcal{S}(\kappa, \tau, T) \iff W_T C_T \leq 1.$$

The two dials enter with different force. The threshold weight leg is PROVED. Its composition leg is NUMERICAL on the order-size-two, $H = 10$, $T = 5$ grid at frozen benchmark policy, for four adjacent threshold pairs and $\kappa \in [0.15, 0.85]$ in steps of 0.01. The clock result is PROVED as an exact equivalence. The model does not sign C_T , so the effect of a shorter clock depends on which histories leave the pooled cell. Because $W_T > 0$, the attenuation criterion is equivalent to $C_T \leq 1/W_T$, and the right-hand side is at least one.

4.4 Who gets caught

Fix the threshold margin τ and the policies, and compare $T' < T$ at a common κ . Write $A = \mathcal{C}_P(\tau, T)$ for the pooled cell under the longer clock and $B = \mathcal{C}_F(\tau, T') \setminus \mathcal{C}_F(\tau, T)$ for the histories the shorter clock newly flags, with $\varphi = \mathbb{P}(B)/\mathbb{P}(A)$ the share of the pool that is caught. Write $s_A = \partial_\kappa \mathbb{E}[h^{(T)} \mid A]$ for the pool's own noise sensitivity in kernel units and s_B for the *caught leg*: the derivative with respect to κ of the premium mass the shorter clock removes from the pool, per unit of caught mass, with the survivors' re-pricing included. It is not the sensitivity of the caught histories at the old prices; the difference is the re-pricing remainder δ , and $s_B = \tilde{s}_B - ((1 - \varphi)/\varphi)\delta$ with $\tilde{s}_B$ that old-price sensitivity.

LABEL: PROVED.

Corollary 1 (Who gets caught; Appendix Section 3.1, Corollary 1). *Under (C-1) to (C-3) of the appendix, with the clock theorem's hypotheses also applying:*

(i) the cut is nested, $B \subseteq A$, $A \setminus B = \mathcal{C}_P(\tau, T')$, and the weight leg is $W_T = 1 - \varphi$;

(ii) the cell masses do not depend on κ , so φ does not either;

(iii) the cut identity holds exactly:

$$s_A = (1 - \varphi) s_{A \setminus B} + \varphi s_B,$$

the longer clock's pooled sensitivity is the mass-weighted average of what the shorter clock leaves in the pool and what it takes out;

(iv) the composition ratio satisfies

$$C_T \leq 1 \iff (s_B - s_A)(\varphi s_B - (2 - \varphi)s_A) \leq 0.$$

Thus $C_T \leq 1$ exactly when s_B lies weakly between s_A and $((2 - \varphi)/\varphi)s_A$. The attenuation criterion $W_T C_T \leq 1$ holds exactly when s_B lies weakly between 0 and $(2/\varphi)s_A$;

(v) *under a common nonzero sign, $C_T \leq 1$ is equivalent to $|s_A| \leq |s_B| \leq ((2 - \varphi)/\varphi)|s_A|$. The caught leg must be at least as noise-sensitive as the pool and no more than that multiple.*

The caught leg turns C_T into a statement about the premium mass removed from the pool. It has two parts: the old-price sensitivity of the newly flagged histories and the re-pricing of the surviving pool. Only when the re-pricing remainder is zero does s_B measure the sensitivity of the caught histories alone.

5 The calibration and the figures

The calibration uses order size two and $H = 10$. A seed policy at $\tau = 0.05$, $T = 5$, and $\kappa = 0.5$ supplies the Voice region. The threshold ladder uses the full-line signal quantiles 0.1, 0.3, 0.5, 0.7, and 0.9 of the stake-at-filing distribution under that seed policy. The calculation clips signals only if a quantile crosses the computational support. The resulting τ values run from 0.0924 to 0.0970. A second policy calculation at the median threshold, $T = 5$, $\kappa = 0.5$, and $H = 10$ gives the benchmark cutoff vector $k = (0.9425, 1.8485)$. Every comparative static freezes this benchmark policy. Expectation calculations use the renormalised computational signal law stated in Section 3.3. The liquidity grid has seventy-one nodes, $\kappa = 0.15$ to 0.85 in steps of 0.01 .

The factorisation on the grid. The largest pointwise residual $|\mathcal{S} - (1 - \Omega)\mathcal{S}_P|$ over the order-size-two, $H = 10$ grid at frozen baseline policies is 5.1×10^{-17} . The flagged weight is flat in κ to exactly 0.0 at every node. Figure 1 draws the median threshold quantile and reports both factors.

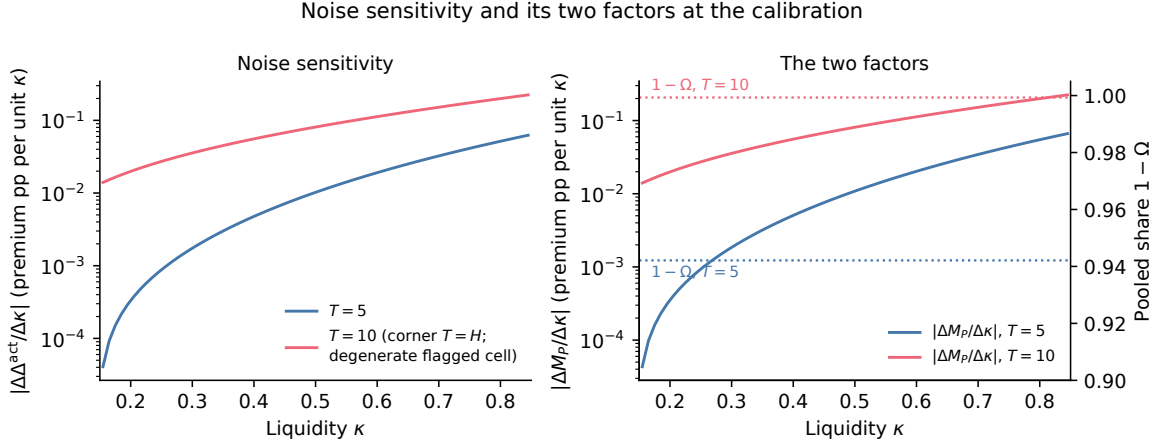


Figure 1: Noise sensitivity and its two factors against liquidity κ at the calibration node (the median threshold quantile, $\tau = 0.0945$), under both clocks, from the order-size-two calibration record at frozen benchmark policy. Sensitivities are slopes over consecutive grid nodes in premium percentage points per unit κ ; the grid is $\kappa = 0.15$ to 0.85 in steps of 0.01 with no gap. Left: $|\Delta\Delta^{\text{act}}/\Delta\kappa|$ (the finite-difference sensitivity). Right: the two factors, the pooled finite-difference sensitivity $|\Delta M_P/\Delta\kappa|$ (left axis, log scale) and the pooled share $1 - \Omega$ (right axis, dotted), the factorisation $|\Delta\Delta^{\text{act}}/\Delta\kappa| = (1 - \Omega)|\Delta M_P/\Delta\kappa|$ holding to machine precision on every node. At $H = 10$ the clock $T = 10$ is the corner $T = H$. Its flagged cell carries mass $\Omega = 6.8 \times 10^{-4}$ at every threshold node, below the code's degenerate-cell floor of 0.01 .

The threshold dial on the grid. The composition ratio is at most one at every adjacent threshold pair in Table 1. This conclusion and the resulting dial inequality are NUMERICAL on the order-size-two, $H = 10$, $T = 5$ grid at frozen benchmark policy, with $\kappa \in [0.15, 0.85]$ in steps of 0.01. At $T = 10$ the ladder reclassifies no mass, so both legs equal one. The weight leg at $T = 5$ lies between 0.975 and 0.977, with reclassified mass 0.0232 at each pair. The composition leg reaches 0.791. The largest value of $W_\tau C_\tau$ is 0.772, for the change from the 0.5-quantile threshold to the 0.3-quantile threshold. The claim covers only the named grid.

Table 1: The threshold dial at fixed policies, from the revelation record (NUMERICAL, grid $\kappa \in [0.15, 0.85]$ in steps of 0.01, order size two, $H = 10$, $T = 5$, frozen benchmark policy). Each row tightens the threshold from the τ quantile in the second column to the tighter one in the first. W_τ is the weight leg; C_τ runs over the whole κ grid; the last column is the largest product on the grid.

Tighter τ quantile	From	Window T	W_τ	C_τ range	$\max_\kappa W_\tau C_\tau$
0.1	0.3	5	0.975	[0.279, 0.456]	0.445
0.3	0.5	5	0.975	[0.219, 0.791]	0.772
0.5	0.7	5	0.976	[0.029, 0.587]	0.573
0.7	0.9	5	0.977	[0.140, 0.633]	0.618

The clock dial on the record. The clock comparison is NUMERICAL on the order-size-two, $H = 10$ grid at frozen benchmark policy, with $\kappa \in [0.15, 0.85]$ in steps of 0.01, five threshold nodes, and compares window margins $T = 5$ against $T = 10 = H$ in discrete trading rounds. This is a model comparison across trading-round horizons rather than a calibrated replica of the statutory reform. The $T = 10$ flagged cell is degenerate at every node, with $\Omega = 6.8 \times 10^{-4}$ below the code's 0.01 floor, so this is a corner-to-interior comparison. Table 2 shows that the shorter clock reduces total-variation sensitivity to between 2 and 69 percent of its long-clock value. The composition ratio runs from 0.024 to 0.699, while the weight leg reduces the pooled share by between 1.1 and 10.4 percent.

Table 2: The window record at fixed policies (NUMERICAL, $\kappa \in [0.15, 0.85]$ in steps of 0.01, order size two, $H = 10$, frozen benchmark policy): cutting the clock from $T = 10$ to $T = 5$ at each threshold node. Sensitivities are total variations of Δ^{act} and M_P over the κ grid. At $H = 10$ the clock $T = 10$ is the corner $T = H$ and its flagged cell is degenerate ($\Omega = 6.8 \times 10^{-4}$, below the code's floor of 0.01), so the comparison is a corner-versus-interior comparison.

τ quantile	$\Omega(\tau, 5)$	W_T	C_T	$W_T C_T$
0.1	0.10420	0.896	0.024	0.021
0.3	0.08104	0.920	0.080	0.074
0.5	0.05789	0.943	0.196	0.184
0.7	0.03473	0.966	0.378	0.365
0.9	0.01158	0.989	0.699	0.691

Who gets caught on the record. On the five-node clock-composition grid, the caught leg lies strictly inside the corollary’s band and $C_T < 1$ at every node (NUMERICAL, frozen benchmark policy, point derivatives at $\kappa = 0.5$, order size two, $H = 10$, $T' = 5$ against $T = 10 = H$, with the $T = 10$ flagged cell degenerate at every node because $\Omega = 6.8 \times 10^{-4} < 0.01$). The pool sensitivity is $s_A = 0.0040$ in kernel units, s_B ranges from 0.039 to 0.135, the upper multiple ranges from 18 to 182, and C_T ranges from 0.013 to 0.643.

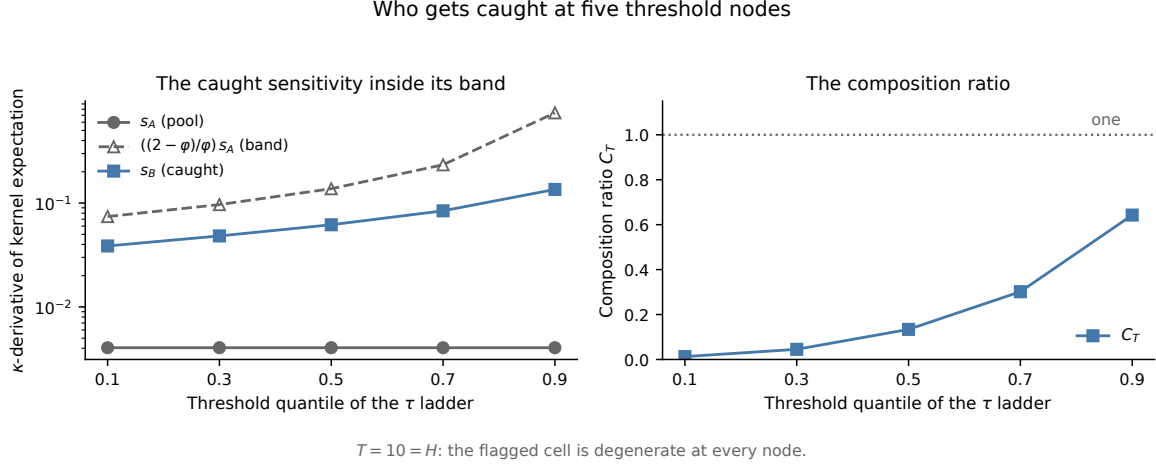


Figure 2: The who-gets-caught record at five threshold nodes. The left panel places s_B strictly between s_A and $((2-\varphi)/\varphi)s_A$, and the right panel has $C_T < 1$, at every node (NUMERICAL, frozen benchmark policy, point derivatives at $\kappa = 0.5$, order size two, $H = 10$, $T' = 5$ against $T = 10 = H$, with the $T = 10$ flagged cell degenerate at every node because $\Omega = 6.8 \times 10^{-4} < 0.01$). The right panel marks 1 as the ceiling.

6 The empirics: the stake at filing and the clock

6.1 Design and gates

The empirical layer is descriptive, reporting filing stakes and delays around the February 2024 rule change to provide institutional magnitudes for the model parameters rather than measuring market inference directly. It identifies no effect, uses no control group, and makes no causal claim. The design fixes the population and measurement before estimation. The measured object is the stake at filing B^F , which the model produces at the filing date.

The population is every initial Schedule 13D filing (form types SC 13D and SCHEDULE 13D, both EDGAR spellings, no amendments) in the EDGAR quarterly form indexes from 2021 Q1 through 2025 Q4, with no screen on Item 4 purpose text. The unit is the campaign, one subject firm and trigger date pair. The procedure collapses simultaneous group filings to the earliest acceptance. The resulting sample has 4914 campaigns. The procedure reads the stake from the cover page as the percent of class and takes the maximum across reporting persons in a joint filing. The period split is the rule date: pre if the trigger predates 2024-02-05, post otherwise. The delay of the clock paragraph is the count of federal business days from the trigger date to the filing date.

The registered quality gates cover parsing, blind hand checks, and differential coverage. Read-

able stakes cover 91.6 percent of campaigns. The blind check reads sixty stratified campaigns and finds zero material stake or trigger-date errors. The pre-to-post coverage gap is 7.0 percentage points. All three gates pass. The following subsection labels the post-minus-pre bootstrap differences ESTIMATED; the registered design treats the remainder of the tables as descriptive statistics.

6.2 The stake at filing

Table 3 reports B^F by calendar year of the filing date and by period; Figure 3 draws the distribution. Among campaigns with a readable stake, the median is 13.1 percent of class before the rule date and 12.8 after, while the mean moves from 22.3 to 23.9.

LABEL: ESTIMATED.

The post-minus-pre difference in the mean is 1.66 percentage points with a campaign-level percentile-bootstrap interval of $[0.20, 3.11]$ based on 2,000 draws and seed 5. The median difference is -0.30 with interval $[-1.62, 1.30]$.

Table 3: The stake at filing B^F , percent of class, by calendar year of the filing date and by period around the 2024-02-05 rule date. Campaigns with a readable stake; the coverage gate E1-G1 passed at 91.6 percent. The difference rows are post minus pre, campaign-level bootstrap, 2,000 draws, seed 5. Descriptive; no control group; no causal claim.

	n	Mean	Median	P25	P75
2021	1040	21.8	14.4	7.7	26.4
2022	807	22.6	12.9	7.0	25.3
2023	856	21.5	11.0	6.7	25.0
2024	843	24.4	12.7	6.8	31.5
2025	956	24.1	13.7	7.2	29.5
Pre (trigger before 2024-02-05)	2888	22.3	13.1	7.2	26.2
Post (trigger from 2024-02-05)	1614	23.9	12.8	6.9	29.4
Difference in mean		1.66 $[0.20, 3.11]$			
Difference in median		-0.30 $[-1.62, 1.30]$			

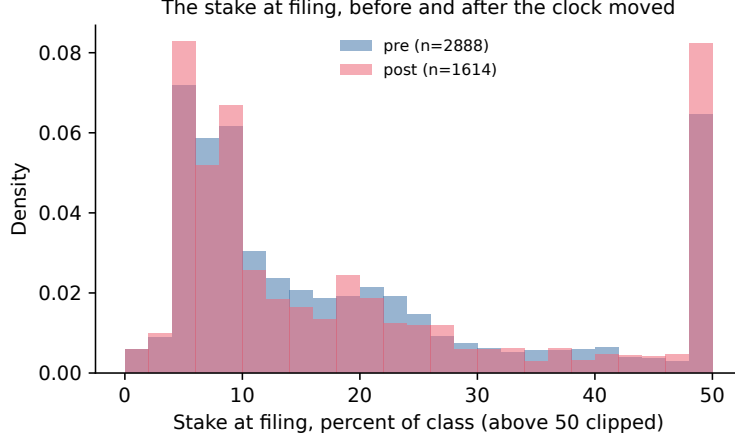


Figure 3: The stake-at-filing distribution by period among campaigns with a readable stake, from the campaign table behind Table 3. The pre period has $n = 2888$ and the post period has $n = 1614$. The medians are 13.1 and 12.8 percent of class. Descriptive; no control group.

By Lemma 9 in the appendix, under fixed policies, a shorter window weakly lowers the stake at filing on every path flagged under both windows. That statement is pathwise. The measured cross-section is not. The shorter clock can bring paths into the filing set that the longer clock left in the pool, so the median and the mean of the filers can move either way. The measured median difference is -0.30 percentage points, while the mean difference is 1.66 points. Table 3 reports their percentile-bootstrap intervals. This comparison is descriptive, and the paper draws no causal conclusion from it. The rule date may coincide with other market changes, so a before-and-after split carries no identification.

6.3 The clock moved

Table 4 documents the change in filing timing. It reports, by calendar year and by period around the rule date, the share of campaigns filed within five business days of the trigger and the median trigger-to-filing delay in federal business days. Before the rule date, 32.7 percent of campaigns were filed within five business days (percentile-bootstrap interval $[31.1, 34.3]$) and the median delay was 7.0 business days; after it, 75.9 percent were filed within five business days ($[73.8, 77.9]$) and the median delay was 5.0. By calendar year, the share within five business days rises from 27.8 percent in 2021 to 39.0 percent in 2023, then to 70.6 percent in 2024 and 70.2 percent in 2025. The median delay across those years is 7.0, 7.0, 6.0, 5.0, and 5.0 days. The statement is descriptive. It documents a change in realised timing at the rule date and identifies no effect on another outcome.

The delay calculation uses all filings with a parsed trigger date and filing date. In the raw records, two filings have a negative delay, reflecting typographical date errors on the cover page, and 138 filings have parsed trigger dates predating 2021 (the earliest reaching back to 2010 with a delay of 2,870 business days), reflecting multi-year legacy accumulations or late disclosures. Retaining these filings preserves the registered descriptive protocol; because the primary statistics are medians and shares within five business days, they are robust to extreme positive or negative delays.

Table 4: The filing clock, by calendar year of the filing date and by period around the 2024-02-05 rule date. The share is campaigns filed within five business days of the trigger; the interval is a campaign-level bootstrap, 2,000 draws, seed 5, reported for the two period rows. Every campaign with a measured delay, no outcome screen. Descriptive.

	Campaigns	Within 5 bus. days, %	Median delay, bus. days
2021	1167	27.8	7.0
2022	903	33.9	7.0
2023	962	39.0	6.0
2024	918	70.6	5.0
2025	964	70.2	5.0
Pre (before 2024-02-05)	3237	32.7 [31.1, 34.3]	7.0
Post (from 2024-02-05)	1677	75.9 [73.8, 77.9]	5.0

7 Conclusion

A disclosure rule partitions the histories that reach a control decision. At fixed plan, cutoff, and execution policies, the expected engagement premium’s liquidity sensitivity equals the pooled share times the pooled cell’s sensitivity. This factorisation and the liquidity invariance of the flagged endpoint are PROVED.

The two legal dials act differently on that partition. The threshold’s weight effect is PROVED. Its composition effect is NUMERICAL on the order-size-two, $H = 10$, $T = 5$ grid at frozen benchmark policy, across four adjacent threshold pairs and $\kappa \in [0.15, 0.85]$ in steps of 0.01. For the clock, the condition $W_T C_T \leq 1$ is a PROVED necessary and sufficient condition for attenuation at fixed policies. The who-gets-caught corollary is also PROVED. It characterises C_T through the sensitivity of the premium mass removed from the pool, including the re-pricing of the survivors. The pooled experiment’s garbling order and polynomial form are PROVED at order size two.

The empirical measurement is descriptive, with the post-minus-pre differences labelled ESTIMATED. Among initial Schedule 13D campaigns with readable stakes from 2021 to 2025, the median stake at filing is 13.1 percent of class before the 2024 clock change and 12.8 percent after. Across all measured delays, the share filed within five business days rises from 32.7 to 75.9 percent, while the median delay falls from 7.0 to 5.0 business days. These measurements describe the stake at filing and the change in realised filing timing at the rule date. They do not identify an effect of the clock on the stake.

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